

Project Tool – Quick Definition

Project lets your sketch reference geometry from elsewhere without copying it. Original geometry becomes purple reference geometry that updates if the source changes.

Core Concept

Original → Project → Purple (linked). You are referencing, not owning the geometry.

Basic Steps

1. Start sketch
2. Press P
3. Select geometry
4. OK

Result: purple reference lines

When to Use Project

- Match size/location exactly
- Align sketches
- Reference faces/edges
- Build from master sketch

What Purple Means

- Linked to source
- Cannot move/edit directly
- Updates automatically
- Can constrain/dimension to

Break Link

Right-click → Break Link
Turns purple into editable geometry (blue/black).

Common Uses

- Copy rectangle location/size
- Align holes to edges
- Use centerlines across sketches

Common Problems

- Can't move → it's linked
- Sketch changes → source changed
- Messy → too much projected geometry

STL Warning

Projecting mesh/STL creates many edges. Use carefully or simplify first.

Project vs Copy

Project = linked reference

Copy = independent geometry

Break Link = convert to editable copy

Best Practices

- Project only what you need
- Use master sketches
- Avoid clutter
- Prefer constraints over copying

Quick Workflow

Hover → Project → Constrain → Build feature